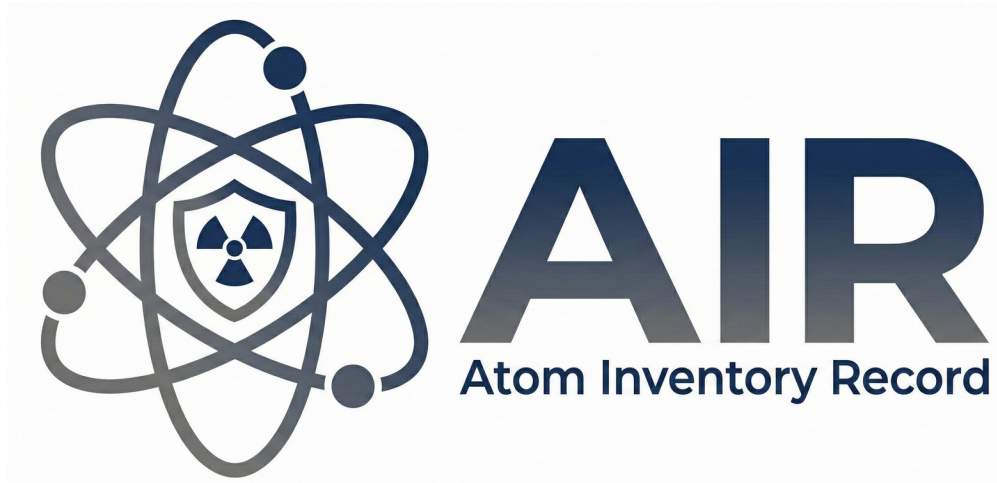




USER MANUAL

Nuclear Material Accountancy and Control System
IAEA Safeguards Compliant

Version 1.0.0

April 2026

Open Source — Educational Tool

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1. Introduction

1.1 Overview

The Atom Inventory Record (AIR) is a secure, cross-platform desktop application designed for Nuclear Material Accountancy and Control (NMAC). It is built to assist nuclear facilities in maintaining accurate, auditable, and IAEA-compliant records of all nuclear material under their custody.

AIR implements the reporting requirements defined in INFCIRC/153 (The Structure and Content of Agreements Between the Agency and States) and supports the preparation of all standard IAEA safeguards reports, including the Inventory Change Report (ICR), List of Inventory Items (LII), Nuclear Loss Items (NLI), Material Balance Report (MBR), and the General Ledger (GL).

1.2 Purpose and Scope

This manual provides complete operational guidance for all authorized users of the AIR system. It covers system login, navigation, data entry, report generation, backup procedures, user administration, and the training simulator.

This manual is intended for:

- Facility Safeguards Officers responsible for IAEA reporting
- Nuclear Material Accountancy operators performing day-to-day data entry
- System Administrators managing user accounts and database backups
- Training coordinators and students using the Training Simulator

1.3 System Philosophy

AIR is built on three core principles:

- Security by Design — Security is embedded into the application logic, not applied as an afterthought. Every critical action is validated, logged, and protected.
- Air-Gapped Operation — AIR requires no external network connection. All data remains on the local machine, minimizing remote attack vectors.
- Tamper-Evident Integrity — Every transaction is recorded with a cryptographic hash chain (SHA-256). Any external modification to the database is immediately detected and visually flagged.

1.4 IAEA Safeguards Compliance

AIR generates reports that comply with the following IAEA documents:

INFCIRC/153	The primary legal basis for comprehensive safeguards agreements between the IAEA and States.
Subsidiary Arrangements (Code 10)	The General Part: detailed procedures for safeguards implementation including reporting formats and timelines.

IAEA-TECDOC series	Technical guidance on nuclear material measurement and accountancy procedures.
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2. System Requirements and Installation

2.1 Supported Platforms

AIR is a native desktop application available on three operating systems:

Platform	Version	Distribution Format
Linux	Ubuntu 22.04+, other distributions	ApplImage (portable, no installation required)
Windows	Windows 10 / Windows 11	.exe installer (automated setup)
macOS	macOS 12 Monterey and later	.dmg disk image (Universal — Intel and Apple Silicon)

2.2 Hardware Requirements

Processor	1.5 GHz dual-core or faster
Memory	4 GB RAM minimum (8 GB recommended)
Storage	500 MB available disk space (plus database growth)
Display	1280 × 800 minimum resolution (1920 × 1080 recommended)
Network	Not required — AIR operates fully air-gapped

2.3 Installation

Linux (ApplImage)

1. Download the AIR ApplImage file to your system.
2. Open a terminal and navigate to the download location.
3. Make the file executable: `chmod +x AIR-1.0.0-x86_64.ApplImage`
4. Run the application: `./AIR-1.0.0-x86_64.ApplImage`

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On first launch, AIR automatically creates the database files in your Documents folder. No installation wizard is required for the ApplImage format.

Windows (.exe Installer)

5. Double-click the AIR_Setup.exe file.
6. Follow the installation wizard. Accept the default installation directory or choose a custom path.
7. Click Install, then Finish when complete.

8. Launch AIR from the Start Menu or the Desktop shortcut.

macOS (.dmg)

9. Double-click the AIR.dmg file to mount the disk image.
10. Drag the AIR application icon to your Applications folder.
11. Open AIR from your Applications folder.
12. If macOS shows a security warning, go to System Preferences > Security & Privacy and click Open Anyway.

NOTE

On macOS, AIR automatically uses the Light theme regardless of system Dark Mode settings to ensure consistent readability of all UI elements.

3. Getting Started

3.1 Intro Screen

When AIR is launched for the first time, the Intro Screen is displayed. It provides an overview of the system, navigation to the Resources section, and links to key documentation including this manual, INFCIRC/153, and the Subsidiary Arrangements (IAEA Code 10).

The Intro Screen has three pages accessible by navigation buttons:

- Welcome — Landing page with the AIR logo and primary action buttons (Enter, Resources, Exit).
- Welcome to AIR — System description and IAEA compliance statement.
- Resources & Documentation — Links to three key reference documents.

At the bottom of the Welcome page, a checkbox labeled "Show this screen each time AIR is started" controls whether the Intro Screen appears on future launches. This setting persists across sessions and can be re-enabled at any time from Administration > Show Intro Screen.

3.2 Login

After the Intro Screen (or directly on subsequent launches if it is disabled), the Login Dialog is displayed.

Login Procedure

13. Enter your Username in the first field.
14. Enter your Password in the second field.
15. Press Enter or click the Login button.

WARNING

If three or more consecutive failed login attempts occur, a CAPTCHA security check is triggered. You must correctly enter the displayed code before the system will process further login attempts. After five failed attempts, the account is temporarily locked for 5 minutes.

Default Administrator Account

On a fresh installation, a default administrator account is created automatically:

Username	admin
Password	Admin@123
Role	Administrator

IMPORTANT

Change the default administrator password immediately after first login. Navigate to

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Administration > User Management to update credentials.

3.3 Main Window Navigation

After successful login, the AIR Main Window is displayed. It consists of two bars at the top and a content area below.

Branding Bar (Top)

The top bar displays the AIR logo and application name on the left, and a Logout button on the right. This bar is always visible regardless of which module is active.

Navigation Bar

Below the branding bar is the Navigation Bar containing four primary navigation items:

Home	Dashboard showing the General Ledger preview and Material Balance preview with live tamper detection.
Operations	Dropdown menu containing all NMAC reporting modules: MBR, ICR, LII, NLI, and General Ledger.
Training Simulator	The safeguards training environment with scenario-based learning modules.
Administration	User management and database backup/restore functions (Administrators only).

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The Administration menu item is only visible to users with the Administrator role. Read-Write and Read-Only users will not see this option.

4. Home Dashboard

4.1 Overview

The Home Dashboard is the default view after login and provides a real-time summary of the facility's nuclear material inventory. It contains two sections: the General Ledger Preview and the Material Balance Preview.

4.2 General Ledger Preview

The upper section of the dashboard displays the General Ledger in IAEA format — a multi-header table showing all inventory changes organized into Increases, Decreases, and running Inventory Balance columns. Each data row is verified against its stored cryptographic hash on every load.

Tamper Detection

If a ledger entry has been modified outside the AIR application (e.g., by direct database manipulation), the affected row is immediately highlighted in red and the IC Code column displays the text TAMPERED. The cell tooltip reads: SECURITY ALERT: Row data altered outside application.

IMPORTANT

A tampered entry does not necessarily indicate a diversion attempt — it could also indicate accidental database corruption. However, it must be investigated and reported to the facility Safeguards Officer immediately.

4.3 Material Balance Preview

The lower section shows the most recent Material Balance Report (MBR) entries, up to a maximum of 10 rows. Each row is also subject to hash chain verification. Tampered MBR entries are highlighted in red with the entry name replaced by TAMPER ALERT.

The Material Balance Preview updates automatically whenever a new entry is added to the MBR or General Ledger from any module.

5. Operations Modules

The Operations menu provides access to all five NMAC reporting modules. Each module follows the same structural layout: a Report Header groupbox at the top, a Data Entry groupbox in the middle, and a Data Table at the bottom.

5.1 Inventory Change Report (ICR)

The ICR module records all changes to nuclear material inventory within the Material Balance Area. Each transaction corresponds to an Inventory Change Document (ICD).

Report Settings

The Report Settings groupbox at the top captures header information used when generating the PDF report:

Country	Searchable dropdown containing all IAEA member states. Type the country name or code to filter.
Report Date	Date of the report. Defaults to today's date. Uses a calendar picker.
Facility	Name of the nuclear facility. Editable text field.
Report No.	Sequential report number. Numeric spinner (1–99999).
MBA	Material Balance Area code. Select from the dropdown or type a new code. Click Save MBA to persist a new code for future sessions.
Shipper/Receiver	Names of the shipping and receiving facilities for transfers.

New Receipt Transaction — Field Reference

Batch Identity	Unique identifier for the material batch (e.g., CRR01). Required field.
Recv. MBA	The MBA receiving the material.
From MBA	The MBA or external source sending the material. Select EXT (External) for foreign receipts.
Inv. Change Code	The IAEA inventory change code (e.g., RD = Receipt Domestic, RF = Receipt Foreign, SD = Shipment Domestic).
No. of Items	Number of physical items in the shipment.
Date	Date of the inventory change. Calendar picker.
Element Code	Material element type: N (Natural U), E (Enriched U), D (Depleted U), P (Plutonium), T (Thorium).
Total Element Wt (g)	Total weight of the element in grams. Precision: 2 decimal places.
Isotope/Fissile Wt (g)	Weight of the fissile isotope (U-235 for uranium, Pu-239 for plutonium). Precision: 3 decimal places.

Material Form	Physical/chemical description of the material (e.g., UO2 Powder, Metal, Solution).
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Adding a Receipt

16. Fill in the Report Settings fields (Country, Facility, MBA, etc.).
17. Enter the Batch Identity (required).
18. Select the appropriate IC Code from the dropdown.
19. Select the Element Code for the material type.
20. Enter the Element Weight and Isotope/Fissile Weight.
21. Enter the Material Form description.
22. Click Add Entry. The entry appears in the table below.

Exporting the ICR to PDF

23. Ensure all Report Settings fields are correctly filled.
24. Click Export PDF.
25. Select a save location in the file dialog.
26. Click Save. A success message confirms the report was generated.

NOTE

The ICR PDF export queries all receipts from the database with change codes RD, RF, and RN. Ensure all entries have been added before exporting.

5.2 List of Inventory Items (LII)

The LII module produces the Physical Inventory Listing — a complete list of all nuclear material items present in an MBA at the time of a Physical Inventory Taking (PIT). This corresponds to the IAEA PIL format.

Adding an Inventory Item

27. Select the Key Measurement Point (KMP) from the dropdown: FFS (Fresh Fuel Store), RRC (Reactor Core), SFS (Spent Fuel Store), or LOF (Location of Facility).
28. Enter the Position (e.g., A01 for rack position) and Batch/Item identifier.
29. Click Select... next to the Mat. Code (430) field to open the Material Description Code Selector.
30. In the Material Description Code Selector, choose the appropriate options from the four categories: Physical Form, Chemical Form, Containment, and Irradiation Status. The 4-character code is generated automatically.
31. Enter the Element weight (g), Fissile weight (g), Plutonium weight (g), and Burnup value.
32. Click Add Entry.

Material Description Code (Label 430)

The Material Description Code is a 4-character IAEA code composed of:

- Character 1 — Physical Form (e.g., F=Powder, B=Fuel Elements, G=Ceramic Powder)
- Character 2 — Chemical Form (e.g., Q=Dioxide, U=U3O8, D=Metal)
- Character 3 — Containment (e.g., 1=Uncontained, 2=Fuel Units, L=Drum 20-50L)
- Character 4 — Irradiation Status (e.g., F=Fresh Fuel, G=Irradiated Fuel, B=Pure Stable)

5.3 Nuclear Loss Items (NLI)

The NLI module records nuclear losses — material that has been used up or lost through nuclear transformation (e.g., fission, neutron capture). Common change codes are LN (Nuclear Loss), LD (Measured Discard), and LA (Accidental Loss).

Adding a Nuclear Loss Entry

33. Enter the Batch Identity for the affected material.
34. Set the number of items and select the Change Code (LN, LD, or LA).
35. For Uranium: enter the Element Code (E, N, or D), Isotope Code, Uranium weight, and Isotope weight.
36. For Plutonium: enter the Plutonium Element Code and Plutonium weight.
37. Click Add Entry.

5.4 General Ledger (GL)

The General Ledger is the master accountancy record for the facility. It records every inventory change in chronological order and maintains a running balance of element and isotope weights. It is the foundation of all IAEA material balance reporting.

Report Information

Facility	Nuclear facility name.
MBA	Material Balance Area. Editable combo with Save MBA functionality.
Material Description	Description of the material type (e.g., Low Enriched Uranium — U3O8-Al).
Element Code	Single letter code for the element (E, N, D, P, T). Printed in the report column headers.
Isotope Code	Single letter code for the isotope (G = U-235, F = Pu-239, etc.).
Unit	Unit of measurement (g = grams, kg = kilograms).

Adding a Transaction

38. Select the Date using the calendar picker.
39. Enter the Reference (ICD/PIL reference number, e.g., ICD-001).
40. Select the IC Code (RF, RD, SD, LN, LD, or type a custom code).

41. Select the Transaction Type from the dropdown.
42. Select the Element type from the dropdown (E=Enriched U, N=Natural U, D=Depleted U, P=Plutonium, T=Thorium). Note: when Plutonium is selected, the Isotope weight field is disabled as it is not applicable.
43. Enter the Element Weight (g) and Isotope Weight (g).
44. Enter the number of Items if applicable (for receipts and shipments).
45. Click Add Entry.

Transaction Types and Their Effect on Balance

Transaction Type	Balance Effect	Column in Ledger
Receipt	Increases balance	Receipts (Increases)
Other Increase	Increases balance	Other (Increases)
Shipment	Decreases balance	Shipments (Decreases)
Other Decrease	Decreases balance	Other (Decreases)
Nuclear Loss	Decreases balance	Other (Decreases)
PIL (Set Balance)	Sets opening balance	Used only for the first entry to establish starting inventory

NOTE

The running balance columns (highlighted in green) are calculated in real-time from all entries. They are not stored independently — deleting an entry recalculates the entire balance from the remaining entries.

5.5 Material Balance Report (MBR)

The Material Balance Report is a periodic IAEA submission summarizing all inventory changes and the resulting material balance for a reporting period. It is required under Article 60 of INFCIRC/153.

Report Details

Country	Searchable IAEA member state dropdown. Exports only the 2-letter country code.
Reporting Period	From and To dates defining the accountancy period.
Facility	Name of the reporting facility.
Report No.	Sequential report number (numeric spinner).
MBA	Material Balance Area. Editable with Save MBA for persistent custom codes.

Accountancy Entry Fields

Entry Name	IAEA accountancy entry code: PB (Physical Beginning), RD (Receipts Domestic), LN (Nuclear Loss), SF (Shipment Foreign), SD (Shipment Domestic), BA (Book Adjustment), PE (Physical Ending), NP (Nuclear Production), MF (MUF).
Element	Element type dropdown: E (Enriched U), N (Natural U), D (Depleted U), P (Plutonium), T (Thorium).
Weight of Element	Element weight. Allows negative values for decreases.
Unit	G (grams) or KG (kilograms).
Fissile Weight (g)	Weight of fissile isotopes. Allows negative values.
Isotope Code	IAEA isotope code dropdown: G (U-235 Enriched), H (U-233), C (U-235 Natural/Depleted), F (Pu-239), J (Pu-241), N (Np-237), T (Th-232), X (Other).
Continuation	Continuation line number for multi-line entries.
Entry Report No.	Report number for this specific entry (numeric spinner).

6. Security Architecture

6.1 Overview

AIR implements a multi-layered security architecture. Security is enforced at the application logic level, not merely at the user interface level. This means that even if a user bypasses the UI, the underlying validation rules still apply.

6.2 Role-Based Access Control (RBAC)

AIR defines three user roles, each with specific permissions:

Role	Access Level	Capabilities
Administrator	Full access	All operations, user management, backup/restore, system configuration
Read-Write	Operational	View and enter data in all operations modules, generate reports
Read-Only	View only	View all data and reports, cannot add, edit, or delete any entries

6.3 Tamper-Evident Hash Chain

Every transaction entered into the system — whether a ledger entry or a material balance entry — is stored alongside a SHA-256 cryptographic hash computed from the record's field values. This hash is recalculated on every load and compared against the stored value.

If the stored hash does not match the recalculated hash, the record is considered tampered. The Home Dashboard immediately displays the affected rows in red with a TAMPERED or TAMPER ALERT indicator. This detection works even if the database file is accessed directly with a third-party SQLite tool.

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The hash chain does not encrypt the data — it detects unauthorized modification. To prevent unauthorized access entirely, ensure the system is physically secured and the database files have appropriate file system permissions.

6.4 Login Security

- Passwords are stored as hashed values. Plaintext passwords are never retained.
- Failed login attempts are counted per session. Three failures trigger a CAPTCHA challenge.
- Five failures trigger a 5-minute account lockout.
- Username input is trimmed to prevent whitespace bypass attempts.

6.5 Zero Trust Verification (Training Mode)

When the Training Simulator is active, all sensitive actions (deleting entries, exporting reports) require a secondary PIN verification. This demonstrates the Zero Trust principle — the system does not trust a single login event for the lifetime of the session, but re-verifies identity for each critical action.

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The Zero Trust PIN in Training Mode is fixed at 1234 for educational purposes. In an operational deployment, this would be replaced with a per-user cryptographic verification mechanism.

7. User Administration

User Administration is accessible only to users with the Administrator role. Navigate to Administration > User Management.

7.1 Viewing Users

The User Management screen displays a table of all registered users showing their ID, username, first name, last name, email address, and role. The default admin account is listed but cannot be deleted.

7.2 Creating a New User

46. Click the + Add new record button to expand the user creation form.
47. Enter the Username (required, must be unique).
48. Select the Rights (role) from the dropdown: Administrator, Read-Write, or Read-Only.
49. Enter the First Name, Last Name, and Email.
50. Enter a Password that meets the complexity requirements:
 - Minimum 6 characters
 - At least one uppercase letter
 - At least one number
 - At least one special character (symbol)
51. The password strength meter displays in real-time: red (weak), orange (acceptable), green (strong).
52. Assign one or more Material Balance Areas (MBAs) to the user using the MBA assignment list.
53. Click Save Record.

NOTE

Assigning MBAs to a user controls which Material Balance Areas they can access and report on. A user with no MBA assignments can still log in but will have limited data visibility.

7.3 Deleting a User

54. Select the user row in the table.
55. Click Delete Selected.
56. Confirm the deletion in the prompt.

NOTE

The default admin account cannot be deleted. Deleting a user also removes all their MBA assignments from the database to prevent orphaned records.

8. Database Backup and Restore

8.1 Overview

AIR includes a built-in backup and restore system to protect against data loss. Backups capture the complete state of the inventory database at a point in time and can be restored if the operational database is corrupted or lost.

IMPORTANT

Backups are stored within the AIR database itself as snapshots. For additional protection, it is strongly recommended to also copy the database files to an external storage medium on a regular schedule.

8.2 Creating a Backup

57. Navigate to Administration > Backup / Restore.
58. Click + Add new record to expand the backup form.
59. Enter a descriptive Title for the backup (e.g., Before_Q1_PIT_2026).
60. Enter an optional Description.
61. Click Update (Create Backup).
62. A success message confirms the backup was created. The new backup appears in the table with its creation date.

8.3 Restoring a Backup

63. Navigate to Administration > Backup / Restore.
64. Select the desired backup in the table.
65. Click Restore Selected.
66. Read the warning carefully. Restoring will overwrite all current data.
67. Click Yes to confirm.
68. A success message is displayed. The application closes automatically to ensure the restored database is loaded cleanly on the next launch.

WARNING

After a restore, relaunch AIR to work with the restored data. The application exits automatically after a successful restore.

8.4 Deleting a Backup

69. Select the backup in the table.
70. Click Delete Selected. The backup is removed immediately without further confirmation.

9. Training Simulator

9.1 Overview

The Training Simulator provides a fully isolated, scenario-based learning environment for students, new inspectors, and early-career NMAC practitioners. All training activities are performed against a separate training database — the operational database is completely unaffected.

The simulator is organized into three difficulty levels with a total of seven training modules:

9.2 Training Modules

Module	Title	Difficulty	Duration	Prerequisites
1.1	The Baseline Receipt	Beginner	30 min	None
1.2	The Reactor Cycle	Beginner	45 min	Module 1.1
2.1	Shipper/Receiver Differences (SRD)	Intermediate	60 min	Level 1
2.2	Physical Inventory Taking (PIT)	Intermediate	75 min	Module 2.1
2.3	MUF Evaluation	Intermediate	90 min	Module 2.2
3.1	Protracted Diversion Detection	Advanced	120 min	All Level 2
3.2	The Substituted Dummy Assembly	Advanced	120 min	All Level 2

9.3 Launching a Scenario

71. Navigate to Training Simulator in the navigation bar.
72. Select the appropriate Level tab (Level 1, Level 2, or Level 3).
73. Review the scenario card including the description, learning objectives, estimated duration, and prerequisites.
74. Click Start Scenario on the desired module.
75. Read the confirmation dialog carefully and click Launch Scenario to proceed.
76. A confirmation dialog indicates Training Mode is now active.

While Training Mode is active:

- The window title changes to AIR — TRAINING SIMULATOR.
- The navigation bar changes to an orange/red color scheme to make the training state visually obvious.
- An Exit Simulation button appears in the navigation bar.
- All data operations work against the isolated training database.

- Sensitive actions (delete, export) require PIN verification (PIN: 1234).

9.4 Exiting Training Mode

77. Click the Exit Simulation button in the navigation bar.
78. Confirm the exit when prompted.
79. The system reconnects to the operational database, the window title reverts, and all widgets refresh with real data.

NOTE

If the application is closed while Training Mode is active, the system automatically reconnects to the operational database on the next launch. Training data is never merged into the operational database.

9.5 Reference Glossary

The Training Simulator includes a built-in Reference Glossary tab with precise definitions for key NMAC terms including MBA, KMP, MUF, ICD, ICR, PIL, SRD, MBR, σ MUF, and INFCIRC/153. Access it by clicking the Reference Glossary tab within the Training Simulator.

10. Troubleshooting

10.1 Common Issues

Issue	Resolution
Cannot log in	Verify username and password. After 3 failed attempts a CAPTCHA is required. After 5 failures the account is locked for 5 minutes.
Intro screen does not appear	Go to Administration > Show Intro Screen to re-enable it. The checkbox controls future launches.
TAMPERED rows on Home Dashboard	A database record has been modified outside AIR. Report to the Safeguards Officer immediately. Do not delete or modify the record without authorization.
PDF export produces empty report	Ensure there is data in the module before exporting. The export requires at least one entry in the table.
MBA not in dropdown	Type the MBA code and click Save MBA to add it permanently. Saved MBAs persist across sessions.
Training Mode stuck after app restart	This should not occur — AIR reconnects to the operational database on every fresh launch. If data appears incorrect, check the window title. If it shows TRAINING SIMULATOR, click Exit Simulation.
Application does not start on macOS	Go to System Preferences > Security & Privacy and click Open Anyway. This is required once for unsigned applications.
Database file not found error	AIR creates database files on first launch in the Documents folder. Ensure you have write permissions to your Documents directory.

10.2 Database File Locations

Linux	~/Documents/air_inventory.db ~/Documents/air_users.db
Windows	%USERPROFILE%\Documents\air_inventory.db air_users.db
macOS	~/Documents/air_inventory.db ~/Documents/air_users.db

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Never delete or move the database files while AIR is running. Always create a backup from within AIR before any manual file operations.

11. Quick Reference — IAEA Change Codes

11.1 Inventory Change Codes

Code	Category	Description
RD	Receipt	Receipt of nuclear material from domestic source
RF	Receipt	Receipt of nuclear material from foreign source
RS	Receipt	Receipt at starting point of safeguards
SD	Shipment	Shipment of nuclear material to domestic destination
SF	Shipment	Shipment of nuclear material to foreign destination
LN	Nuclear Loss	Nuclear loss (fission, neutron capture, etc.)
LD	Discard	Measured discard (waste)
LA	Accidental Loss	Accidental loss of nuclear material
NP	Production	Nuclear production (Pu-239 from neutron capture in U-238)
BA	Adjustment	Book adjustment — correction of a previously reported error
EU	Exemption	Exemption from safeguards based on use
EQ	Exemption	Exemption from safeguards based on quantity
TU	Termination	Termination of safeguards application

11.2 Element Codes

E	Enriched Uranium — U-235 enrichment greater than 0.711%
N	Natural Uranium — U-235 enrichment at 0.711%
D	Depleted Uranium — U-235 enrichment less than 0.711%
P	Plutonium — all isotopes of Pu
T	Thorium — Th-232

11.3 Isotope Codes

G	U-235 (Enriched Uranium)
C	U-235 (Natural or Depleted Uranium)
H	U-233
F	Pu-239
J	Pu-241

N	Np-237
T	Th-232
X	Other isotope

Appendix A — Keyboard Shortcuts and Tips

Enter / Return	Submit the login form or activate the focused button.
Tab	Move between input fields in all forms.
Esc	Close dialogs and cancel operations.
Calendar icon	Click the calendar icon on any date field to open a date picker.
Save MBA	Type a custom MBA code then click Save MBA to persist it permanently.
Select... (Mat.Code)	Opens the 4-character Material Description Code builder dialog for IAEA Label 430.

Appendix B — Acronyms and Definitions

AIR	Atom Inventory Record — the application described in this manual.
ICD	Inventory Change Document — a document recording a change in nuclear material quantity.
ICR	Inventory Change Report — the IAEA submission based on ICDs.
IAEA	International Atomic Energy Agency.
INFCIRC/153	IAEA document defining the structure of comprehensive safeguards agreements.
KMP	Key Measurement Point — a location where nuclear material is measured.
LII	List of Inventory Items — the IAEA physical inventory listing (PIL).
MBA	Material Balance Area — a defined area for nuclear material accountancy.
MBR	Material Balance Report — periodic IAEA submission of material balance.
MUF	Material Unaccounted For — difference between book and physical inventory.
NLI	Nuclear Loss Items — the IAEA report of nuclear losses.
NMAC	Nuclear Material Accountancy and Control.
PIL	Physical Inventory Listing — see LII.
PIT	Physical Inventory Taking — the physical counting of all nuclear material.

SRD	Shipper/Receiver Difference — discrepancy between declared and measured quantities.
σMUF	Standard uncertainty of MUF — used to assess statistical significance of discrepancies.

END OF DOCUMENT

AIR — Atom Inventory Record | User Manual | Version 1.0.0
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